Sơ Đồ Lớp Backend: Các Dịch Vụ (Services) & Thực Thể Dữ Liệu (Prisma Entities)

Backend Services (NestJS Providers)
PaymentsService
+createPayment(userId: string, key: string, body: CreatePaymentDto, mockOutcome?: string): Promise<PaymentResponse>
-replayExistingPayment(payment, userId: string, requestHash: string): PaymentResponse
CancelOrderService
+cancel(orderId: string, userId: string): Promise<CancelResult>
MockGatewayService
+processPayment(mockOutcome?: string): 'SUCCESS' | 'FAILED'
LoyaltyService
+award(tx, order: { id, userId, finalAmount }): Promise<void>
HistoryService
+findAll(userId: string, query?: GetOrderHistoryDto): Promise<OrderHistoryResult>
+findOne(userId: string, orderId: string): Promise<OrderDetail>
AuthService
+register(dto: RegisterDto): Promise<AuthResult>
+login(dto: LoginDto): Promise<AuthResult>
ProductDetailService
+getProductById(id: string): Promise<ProductDetailDto>
OrdersService
+create(userId: string, dto: CreateOrderDto): Promise<Order>
ProductsService
+getProducts(): Promise<ProductItem[]>
OrderService
+transition(tx, orderId: string, toStatus: OrderStatus, meta): Promise<Order>
PrismaService
+onModuleInit(): Promise<void>
+onModuleDestroy(): Promise<void>
+$transaction(fn): Promise<any>
PromotionsService
+validate(code: string, subtotal: number): Promise<PromotionValidationResult>
PricingService
+calculate(items: PricingItemInput[]): Promise<PricingCalculationResult>
processPayment()
award loyalty points
reads/writes payment records
restocks inventory
transition(PENDING -> PAID/FAILED)
transition(CANCELLED)
uses transaction & lock
delegates coupon validation
delegates pricing
creates & queries
queries detail & toppings
creates Order & OrderItems
checks & decrements stock
queries active products
logs state transitions
records payment transaction
Domain Entities (Prisma Models)
«model»
User
+id : String [PK]
+email : String [UK]
+passwordHash : String
+fullName : String?
+loyaltyPoints : Int
+createdAt : DateTime
+updatedAt : DateTime
«model»
Promotion
+id : String [PK]
+code : String [UK]
+description : String?
+discountPercent : Int?
+startDate : DateTime
+endDate : DateTime?
+isActive : Boolean
+createdAt : DateTime
+updatedAt : DateTime
+discountAmount : Decimal(12,0)?
+maxDiscount : Decimal(12,0)?
+minOrderAmount : Decimal(12,0)
«model»
Category
+id : String [PK]
+name : String [UK]
+slug : String [UK]
+displayOrder : Int
+createdAt : DateTime
+updatedAt : DateTime
«model»
Order
+id : String [PK]
+orderNumber : Int [UK]
+userId : String [FK]
+promotionId : String? [FK]
+status : OrderStatus
+pointsEarned : Int
+paidAt : DateTime?
+version : Int «StateLock»
+createdAt : DateTime
+updatedAt : DateTime
+totalAmount : Decimal(12,0)
+discountAmount : Decimal(12,0)
+finalAmount : Decimal(12,0)
«model»
Product
+id : String [PK]
+categoryId : String? [FK]
+name : String
+imageUrl : String?
+stock : Int
+version : Int «OptimisticLock»
+isAvailable : Boolean
+createdAt : DateTime
+updatedAt : DateTime
+basePrice : Decimal(12,0)
«model»
LoyaltyTransaction
+id : String [PK]
+userId : String [FK]
+orderId : String? [FK]
+points : Int
+type : LoyaltyTransactionType
+reason : String
+createdAt : DateTime
«model»
OrderItem
+id : String [PK]
+orderId : String [FK]
+productId : String [FK]
+size : ProductSize
+quantity : Int
+unitPrice : Decimal(12,0)
+lineTotal : Decimal(12,0)
«model»
OrderStatusHistory
+id : String [PK]
+orderId : String [FK]
+fromStatus : OrderStatus?
+toStatus : OrderStatus
+changedBy : String?
+reason : String?
+createdAt : DateTime
«model»
ProductSizePrice
+id : String [PK]
+productId : String [FK]
+size : ProductSize (S|M|L)
+priceAdjustment : Decimal(12,0)
«model»
Payment
+id : String [PK]
+orderId : String [FK]
+idempotencyKey : String [UK]
+requestHash : String?
+method : PaymentMethod
+status : PaymentStatus
+failureReason : String?
+transactionRef : String?
+createdAt : DateTime
+amount : Decimal(12,0)
«model»
Topping
+id : String [PK]
+name : String [UK]
+stock : Int
+isAvailable : Boolean
+createdAt : DateTime
+updatedAt : DateTime
+price : Decimal(12,0)
«model»
OrderItemTopping
+id : String [PK]
+orderItemId : String [FK]
+toppingId : String [FK]
+price : Decimal(12,0)
1
0..*
0..1
1..*